

Final Project

MTH 9899 Baruch College
DATA SCIENCE II: Machine Learning

March 11, 2024

- For the project, work within the groups assigned
- Do not use any data other than the files provided.
- Your code must work as-is, more details provided below

Goals The goal of this project is to go through the process of applying ML techniques that we have been discussing in class to a real-world dataset. The hope is that you will apply the techniques you have been taught in new/interesting ways to predict future market returns. In pursuit of this goal you are encouraged to:

- Come up with useful features from the raw data
- Use cross-validation to fit and choose models
- Tune hyperparameters the best you can
- Use best practices with regards to data cleaning
- Avoid using features that are common across stocks that your model may over-fit to but you are unlikely to find useful out-of-sample (e.g. date)
- Time permitting, implement and test multiple ideas from the semester, and combine them to produce superior results to what any single algorithm can do

Your model will be tested on a hold-out sample to find out how well it performs.

Data You have been given 5 years of **daily** US stock market data covering a liquid universe of approximately 500 stocks. Please refer to the Daily Data Addendum doc for details on the data.

In addition you have been provided **intraday** data at 15 minute intervals consisting of the following fields

- *Date*, Date of the data (same as file date)
- *Time*, snapshots at 26 times are provided, the first at 9:45 and the last at 16:00
- *Id*, a unique identifier for each stock
- *CumReturnResid*, residual return (i.e. after removing factor returns) for that stock from the prior day's close (at 16:00) to "now"
- *CumReturnResidRaw*, the raw return for that stock for the same period.

Notes:

- Returns are cumulative i.e. the return at 10:00 includes the return at 09:45.
- The first snapshot (i.e. at 09:45) contains the return from 16:00 the previous day, through 09:45 that morning.
- Returns provided are adjusted for corporate actions.
- *CumVolume*, cumulative volume for the stock, in shares so far "today"

Some notes about the data:

- The set of stocks will vary slightly from day to day as names fall in and out of the eligible universe. Be mindful of survivorship bias.
- Be mindful of weekends/holidays when creating your features.

Performance metric We will be using out-of-sample weighted R^2 of your predicted forward residual return vs. the actual forward residual return as the final benchmark.

We are preserving a hold-out test dataset (1 year of data). This dataset will have the same raw input fields as the training dataset. You will be provided this data 1-2 days prior to the submission deadline. **Make sure you write your code in the prescribed manner such that you can quickly compute features and apply your previously trained learner to make predictions - DO NOT REFIT YOUR MODEL AT ALL AFTER THE NEW DATA IS PROVIDED.**

Since there may be different ways to calculate weighted r-squared, please use scikit-learn's `r2_score` implementation. Compute weighted R^2 , across dates and IDs, using $\sqrt{MDV_63}$ as weights.

Clip actual returns at 5 MAD cross-sectionally before measuring OOS performance.

Target variable Make predictions at 15:30 for the return over the next 24 hrs i.e. use the next day's cumulative residual return at 15:30, combined with the return from 15:30 to 16:00 "today" as your target variable. Consider clipping your target variable prior to fitting.

We also recommend normalizing your return (with *EST_VOL*) prior to fitting. If you do so, after you make predictions, make sure to scale your prediction back up by the then-current *EST_VOL* to convert your prediction back into "return space".

Sample Weights Use something sensible as weights for your fits, e.g. *MDV_63*. Consider using `sqrt()` or `log()` to mute the effect of a few IDs dominating your fit. Explain why we might want to use something like *MDV* as weights for the fit.

Validation data Do all of your feature engineering and testing using only the first 4 years of data. Keep the last year of data out of sample from yourself. After you have completed your research (i.e. you have made choices for your features, model type, hyper-parameters, etc.), apply your model to the 2014 data and **report (in your white paper) the out of sample R^2 , weighted by $\sqrt{MDV_63}$.**

After having measured and reported your R^2 on this validation dataset, you may choose to re-train your model on data including 2014, for use by us on the test dataset.

Code and model You must submit:

1. Python code, including a "main.py" which can be invoked from the command line and takes the following cmd line options (we recommend using *argparse*):
 - -i: input directory (used differently in Mode 1 and Mode 2, described below)
 - -o: output directory (used differently in Mode 1 and Mode 2)
 - -p: directory containing the learned model/s you provided (only for Mode 2)
 - -s: start date in YYYYMMDD format
 - -e: end date in YYYYMMDD format
 - -m: Mode you will run in. You should support two modes, defined below
2. Your fitted model object/s, in pickle form. The pickle file/s will be stored in a directory which we will point to in Mode 2 above.

Modes As mentioned above, your code should run in 2 modes:

Mode 1: In this mode, you only create features between startdate and enddate (i.e. features used for prediction by your model at 15:30 on the date specified) and save to the output directory specified, a CSV file per date with features for each ID. In this mode "-i" specifies the location of the raw input data (i.e. within this location will be the directories: return, price_volume, etc.) and "-o" specifies where to save your feature files.

We will test for leakage by applying this mode with and without input directories containing future raw data and expect the features to be identical. If your feature creation is found to suffer from leakage, your out-of-sample R^2 will be invalidated and points will be deducted.

Mode 2: In this mode, you are to make predictions between startdate and enddate by applying the learned model you provided to features produced from a run in Mode 1. In this mode "-i" specifies the location of the features previously created and "-o" specifies where to save your predictions. Your predictions should be saved as CSV files, one file per date, with the columns: Date, Time, Id and Pred.

White Paper Each group will submit a detailed white paper detailing your process, models and methodology, including but not limited to:

- How you split the data to create/test features, chose hyperparameters, etc.
- Categories of features created (e.g. time series based, cross-sectional) and full list
- How you tested your features and any cleaning/normalization
- Feature importance, using a feature-permutation method like MDA/MDI
- Include bin plots of your most significant features vs. the target variable
- Choice you made for weights during fitting and why
- Techniques attempted, how you evaluated them and how each performed
- Hyperparameter choices attempted, how you evaluated them and how you made choices (if you plot error curves, please include error bars).
- Performance on the validation dataset (i.e. 2014)
 - R^2 , weighted by $\frac{1}{estVol}$
 - Include a "bin plot" as described in the *Visualization* section below
 - Include a "drift plot" as described in the *Visualization* section below
 - Include a plot showing a 30-day moving avg of correlation (of your predictions vs. the target variable, cross sectionally)
 - Feature importance, using MDA (Mean decrease in accuracy)

Visualization Create the following for each of the 5 years of data provided, using the model fit on the training data only (i.e. plots will be in-sample for the first 4 years, and one of the plots will be out-of-sample):

- A histogram showing the distribution of your predictions.
- A bin plot, similar to ones we looked at in class, with your predictions binned into 4 bins on the X-axis and the actual target value on the Y-axis.
- A drift plot consisting of one plot, with 4 lines, created as follows:
 - Take all of your predictions for the year being plotted and bin them into **4 bins** (using something like *pandas.qcut*)
 - Draw a series for each of these 4 bins showing the actual cumulative residual return (on average, across all predictions in the bin) corresponding to the predictions from 24 hours prior to prediction time, until 48 hours after prediction time, using the 15 minute interval returns to create this trajectory.
 - Note that each series will have 74 points (26 intervals back and 52 intervals forward).
 - Your x-axis can be minutes or an interval index but make sure to mark the prediction time as with negative numbers for points going "back" and positive numbers going "forward". Draw a vertical line at $x=0$.
 - Compute a standard error around the mean (within each bin and for each interval) and plot a dotted line on either side of each series, 1 std. error away.

Grading Some attributes that will contribute to your grade:

- i The process you followed to avoid overfitting.
- ii Creativity of your features
- iii Thoughtfulness of your features and process followed to vet individual features (e.g. just trying random combinations and powers will not be considered as well thought-out features).
- iv Steps taken to normalize your features and/or ensure their stationarity
- v Steps you may have taken to avoid learning spurious signals or that may make you take inadvertent risks (e.g. a strategy that goes long higher volatility stocks and short lower volatility stocks, or one that is size-biased).
- vi Thoughtfulness of your model selection and tuning process.
- vii How your model's performance holds up out-of-sample (both in absolute terms and how close it is to your performance on the validation data).
- viii Visualizations - particularly, in their ability to convey information (the information itself doesn't have to be a positive outcome).